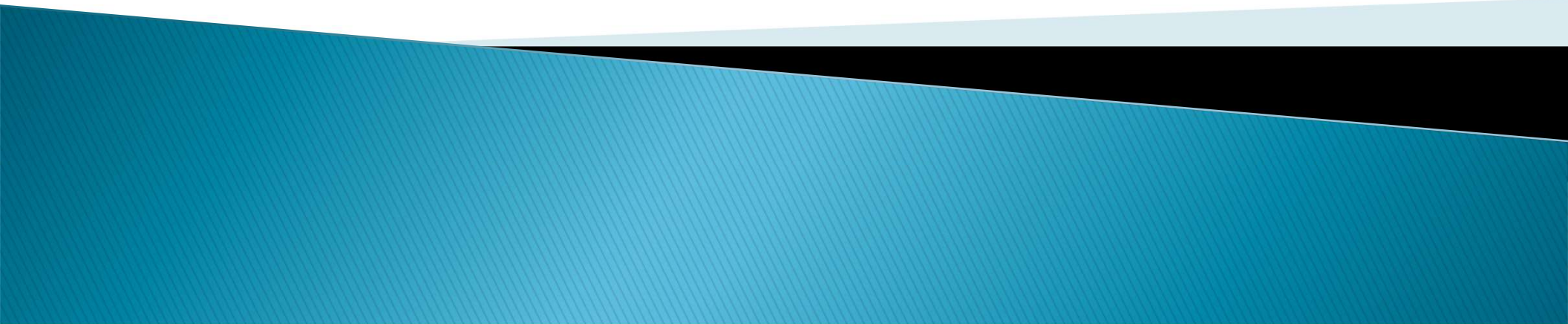


The gaseous states



We shall learn...

the postulates of kinetic molecular theory of gases

Define ideal and real gases

The difference between real and ideal gas

Deduce gas laws from the kinetic gas equation

Find the value of R where R is a gas constant

Mathematical problem



the postulates of kinetic molecular theory of gases

1. A gas consists of extremely small discrete particles called molecules dispersed throughout the container.
 2. Gas molecules are in constant random motion with high velocities.
 3. The gas molecules can move freely, independent of each other.
 4. There is no loss of the kinetic energy of a molecule during a collision.
 5. The pressure of a gas is caused by the hits recorded by molecules on the walls of the container.
 6. The average kinetic energy ($\frac{1}{2} m\bar{v}^2$) of the gas molecules is directly proportional to absolute temperature
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Define ideal and real gases

Ideal gas

- The gas which obeys molar gas equation $PV = nRT$, in all temperatures and pressure.

- It should follow the equation

$$\left(\frac{\partial E}{\partial V}\right)_T = 0$$

- In reality there is no existence of ideal gas

Real gas

- The gas which does not follows any gas laws at any temperature and pressure.

Example: Oxygen, nitrogen, hydrogen etc.



The difference between real and ideal gas

	Ideal Gas		Real Gase
1	The gas which follows "gas laws" such as Boyle's, Charles' and Avogadro's law is known as "ideal gas".	1	The gas, which does not follow gas laws at any temperature and pressure, is known as "real gas".
2	Intermolecular attraction force of ideal gas is negligible.	2	Intermolecular force of real gas is considerable but not negligible.
3	The volume of the molecules is negligible compared to the volume of the container.	3	Volume of the gas molecule is considerable but not negligible.
4	Ideal gas follows gas laws at any temperature and pressure.	4	Real gas follows gas laws at high temperature and low pressure.
5	Ideal gas is an imaginary concept. There is no example of ideal gas.	5	There are many examples of real gas such as H_2 , N_2 , CO_2 etc.

Establish the ideal gas equation

$$PV = nRT$$

From Boyle's law , $V \propto \frac{1}{P}$, V for volume and P for pressure

From Charles law , $V \propto T$, T for temperature and

From the Avogadro's law, $V \propto n$, n for mole number

When we combine the above three laws

$PV \propto nT$ or $PV = nRT$, where R is gas constant.

This equation is called ideal gas equation.



Deduce gas laws from the kinetic gas equation

(a) Boyle's Law

According to the Kinetic Theory, there is a direct proportionality between absolute temperature and average kinetic energy of the molecules *i.e.*,

$$\frac{1}{2}mNu^2 \propto T$$

or
$$\frac{1}{2}mNu^2 = kT$$

or
$$\frac{3}{2} \times \frac{1}{3}mNu^2 = kT$$

or
$$\frac{1}{3}mNu^2 = \frac{2}{3}kT$$

Substituting the above value in the kinetic gas equation $PV = \frac{1}{3}mNu^2$, we have

$$PV = \frac{2}{3}kT$$

For a definite temperature $\frac{2}{3}kT$ is constant

Thereby, $V \propto \frac{1}{P}$

Deduction of Charles's law from the kinetic gas equation

(b) Charles' Law

As derived above,

$$PV = \frac{2}{3}kT$$

or

$$V = \frac{2}{3} \times \frac{k}{P} T$$

At constant pressure,

$$V = k' T$$

$$\text{where } \left(k' = \frac{2}{3} \times \frac{k}{P} \right)$$

or

$$V \propto T$$

That is, at constant pressure, volume of a gas is proportional to Kelvin temperature and this is Charles' Law.

Deduction of Avogadro's law from the kinetic gas equation

(c) Avogadro's Law

If equal volume of two gases be considered at the same pressure,

$$PV = \frac{1}{3}m_1N_1u_1^2 \quad \dots \text{Kinetic equation as applied to one gas}$$

$$PV = \frac{1}{3}m_2N_2u_2^2 \quad \dots \text{Kinetic equation as applied to 2nd gas}$$

$$\therefore \frac{1}{3}m_1N_1u_1^2 = \frac{1}{3}m_2N_2u_2^2 \quad \dots(1)$$

When the temperature (T) of both the gases is the same, their mean kinetic energy per molecule will also be the same.

$$\text{i.e.,} \quad \frac{1}{3}m_1u_1^2 = \frac{1}{3}m_2u_2^2 \quad \dots(2)$$

Dividing (1) by (2), we have

$$N_1 = N_2$$

Or, under the same conditions of temperature and pressure, equal volumes of the two gases contain the same number of molecules. This is Avogadro's Law.

Find the value of R where R is a gas constant

We know the ideal gas equation is $PV = nRT$ where

P - pressure, V – volume, n - number of moles

R - Universal Gas Constant, T – temperature

We can rearrange the Ideal Gas Law to get $R = PV/nT$

$$R = \frac{PV}{nT} = \frac{1.013325 \times 10^5 \text{ Pa} \times 22.414 \times 10^{-3} \text{ m}^3}{1 \text{ mol} \times 273.16 \text{ K}}$$

$$= 8.3145 \text{ Pa} \cdot \text{m}^3 \text{K}^{-1} \text{mol}^{-1}$$

Calculate the root mean square velocity of N_2 molecule at 27°C

We know that

$$\begin{aligned}C &= \sqrt{\frac{3RT}{M}} \\&= \sqrt{\frac{3 \times 8.316 \times 10^7 \times 300}{32}} \\&= 4.8362 \times 10^4 \text{ cm s}^{-1} \\&= 483.62 \text{ ms}^{-1}\end{aligned}$$

Ans. $C = 483.62 \text{ ms}^{-1}$

Here,

Temperature, $T = 27 + 273 = 300 \text{ K}$

Molar mass of O_2 , $M = 32$

Gas constant value

$$= 8.316 \times 10^7 \text{ erg K}^{-1} \text{ mol}^{-1}$$

Root mean square velocity, $C = ?$

Thank you
all

